AI Research & Document Assistant

Project Brief: idea, problem, features, and first-version scope

1. Project Idea & Objective

The project is a web chat assistant that helps a team work with a library of research papers. Team members upload PDF papers, then ask questions, request summaries, and ask for report drafts. Every answer is grounded in the papers and shows its source as a file name and page number. It runs entirely on a local machine with a free, open-source language model, so there are no API costs.

Objectives

- Cut the time the team spends reading and searching through papers.
- Give trustworthy answers that can be checked, with a citation on every claim.
- Support both Arabic and English, in documents and in questions.
- Stay free to run and simple to set up on a modest local machine.
- Serve as a practical first project for learning agentic AI (an LLM that chooses tools).

2. Problem Statement

Teams that follow research read many papers, and the knowledge inside them is hard to reuse. Finding the right paper, locating the relevant passage, and writing a summary by hand is slow and repetitive. The problem is sharper for teams that work in both Arabic and English, because most search tools handle one language well and the other poorly.

General chatbots can answer quickly, but they may invent facts or cite papers that do not exist, and they do not know the team's own paper collection. Paid cloud tools add recurring cost. The team needs an assistant that answers **only from its own papers**, **shows exactly where each answer came from**, works in **both languages**, and runs **locally at no cost**.

Pain point	Impact
Slow manual reading	Hundreds of papers cannot be read or searched by hand in reasonable time.
Untrusted answers	General AI tools may hallucinate and give no way to verify a claim.
Language gap	Arabic and English content is rarely handled well in the same tool.
Cost and setup	Paid APIs add cost, and the team wants a free local solution.

3. Main Features

Feature	Description
Paper library	Add PDF papers (hundreds) to a searchable library. Text is extracted page by page so page numbers are kept.
Question answering with citations	Ask a question in Arabic or English and get an answer built from the papers, each claim tagged with file name and page number.
Document summarization	Summarize a single paper into a short, structured overview.
Report drafting	Ask for a report on a topic. The assistant searches several papers and drafts it in the chat, with citations.
Bilingual support	Understands Arabic, English, and mixed questions, and replies in the language of the question.
Source-only answers	If the papers do not contain the answer, the assistant says so instead of guessing.
Web chat interface	A simple browser-based chat that the whole team can open.
Free and local	Runs on a local machine with an open-source model. No paid API required.

How the agent works: the language model reads the question, decides which tool to call (search the papers, summarize a paper, or write a report), reads the result, and then writes the final answer with citations.

4. MVP Scope

ما سيتم تنفيذه في النسخة الأولى

The first version should be small enough to finish and test, while proving the core idea.

Included in the MVP	Left for later versions
• Load PDF papers from a folder and build a searchable index. • Chat interface in the browser. • Question answering with file name + page number citations. • Summarize one selected paper. • Draft a short report in the chat from a topic request. • Arabic and English questions and answers. • Local open-source model that fits the machine. • Clear "not found in the papers" reply when there is no support.	• Exporting reports as Word or PDF files. • Confidential or private documents. • Web search or automatic paper download. • OCR for scanned PDFs. • User accounts and permissions. • Advanced analytics or paper comparison dashboards.

MVP success check

- Citations point to the correct paper and page in most test questions.
- Both Arabic and English questions return useful answers.
- Answers arrive in a reasonable time on the local machine.